

## **AERO:**

### **Elicitation:**

#### Questions:

1. When do you use this kind of quiz tool the most?
2. What do you find challenging when using current quiz tools for learning?
3. How important is real-time feedback during a quiz session?
4. Are there any features you wish these quiz tools had that would improve your learning?

### **Interviewees:**

Name: Vania

Status: Postgraduate student

Email: vaniiiiiaaa14@gmail.com

1. In tutorials, quizzes are often used by my tutor to measure the understanding of the class.
2. I feel like there's limited customization possibilities, it doesn't give us enough flexibility to create different question types and formats. In most quiz tools, it only allows us to create multiple choice questions, where sometimes it is not enough to meet certain learning objectives.
3. It is very important to me, as it allows me to directly understand the mistakes that I made. Real-time feedback would allow me to directly ask for any clarifications from my tutor. And also, I think real-time feedback just makes it more engaging for everyone to participate.
4. I really wish that it could provide detailed feedback on every question that I got wrong. As it would speed up my learning process, letting me to understand what I did wrong.

Name: Mahta

Status: Undergraduate Student

Email: mahta.mir2003@gmail.com

1. I have used it during both lectures and tutorials to review key concepts.
2. One challenge is the limited time for each question, which often creates pressure and distracts from learning. Sometimes, the fast pace feels more competitive than educational.
3. Real-time feedback allows me to correct misunderstandings immediately and maintains my focus during sessions.
4. It would be great if they included more in-depth explanations for each answer to enhance understanding. Also, it would be really helpful if they allowed for adjustable time limits on questions, giving us more time to think critically.

Name: Alex

Status: Undergraduate Student

Email: alexrubica1997@gmail.com

1. I typically use quiz tools most often during lectures, especially when I need to test my understanding of the material being taught.
2. One of the main challenges I face is the pressure of time limits. When quizzes are timed, it can create unnecessary stress, and I sometimes feel rushed to answer questions.
3. Real-time feedback is very important to me. It helps me understand what I got wrong right away, which ensures that I don't forget important details.
4. One feature that I wish quiz tools had would be flashcards after the quiz. This would allow me to revisit the questions I struggled with and further solidify my understanding.

Name: Bonnie

Status: Undergraduate Student/ Maths Tutor

Email: [libonnie6@gmail.com](mailto:libonnie6@gmail.com)

1. When testing my students on large pieces of info as a summary and recapping a long lesson or just a fun interactive tool.
2. It's hard to type math equations in questions and answers without it looking confusing such as adding indices.
3. Very important as it shows room for improvement and what we got wrong.
4. Explanation of the answers.

**Proposed solutions:**

1. Customization for question types: Allow users to create different types of questions beyond just multiple-choice, such as short answer, true/false, or matching types. This would address the need for flexibility in quizzes.
2. Adjustable time limits: Include an option to adjust time limits per question, so that users can have more time to think critically on difficult questions.
3. Real-time feedback: Implement real-time feedback that provides immediate responses after each question, including detailed explanations to clarify mistakes.
4. In-depth answer explanations: Offer thorough explanations for every incorrect answer to speed up learning and comprehension
5. Flashcard Integration Post-Quiz: Integrate a flashcard feature that automatically generates cards based on the learner's incorrect answers. After completing the quiz, the tool could generate a personalized set of flashcards, focusing on areas where the learner struggled. This will help them in their learning processes.

## **Analysis and Specification - Use Cases:**

### **User Stories**

- User acceptance criteria

Use Case describing solution that satisfies some of or all elicited requirements.

### **User Story 1: Customizable Question Types**

As a quiz creator, I want to create a variety of question types, such as matching, true/false, and short answer, so that I can create more diverse quizzes and assess students better.

#### **Acceptance Criteria:**

1. The quiz creator can choose from at least three different question types: matching types, short answer, true/false, drag and drop.
2. The system validates and saves questions with different formats successfully.
3. When creating a quiz, the creator can see a list of available question types to select from.
4. The system allows the creator to set different options for each question type (e.g., for drag and drop, the creator can add answer choices, and define drop zones).

#### **Steps:**

1. Quiz creator logs into the system.
2. Quiz creator selects the quiz type and starts creating questions.
3. The system allows the creator to choose the question type (multiple choice, short answer, true/false, matching, drag and drop).
4. The system validates the inputs and saves the quiz.

## **User Story 2: Adjustable Time Limits for Questions**

As a quiz creator, I want to set custom time limits for each question, so that students can spend more time on challenging questions and less on easier ones.

### **Acceptance Criteria:**

1. The quiz creator can set a time limit for each individual question.
2. When taking the quiz, students can see how much time is remaining for each question.
3. The system enforces the time limit for each question and automatically moves to the next question once time runs out.
4. The system allows the quiz creator to set different time limits for different questions.

### **Steps:**

1. Quiz creator sets different time limits for each question during quiz creation.
2. The system allows the quiz creator to set specific time limits for each question.
3. During the quiz, students see the adjusted time limit per question.

## **Use Story 3: Real-Time In-Depth Feedback**

As a student, I want to receive immediate, in-depth explanations after answering a question, so that I can learn from my mistakes right away and understand why my answer was wrong, which helps me grasp key concepts more effectively.

### **Acceptance Criteria:**

1. The system provides a detailed explanation of each question immediately after the student submits their answer.
2. If the student's answer is incorrect, the explanation should clearly explain why the answer is wrong and highlight the key concepts behind the correct answer.

3. The explanation can include text, images, or links to additional resources to reinforce learning.
4. Feedback is personalized based on the student's performance, providing suggestions for areas to review or practice further.

**Steps:**

1. Student starts taking the quiz.
2. After answering each question, the system provides immediate feedback on their answer.
3. The system includes a detailed explanation for both correct and incorrect answers.

**User Story 4: Flashcard Review Post-Quiz**

As a student, I want to receive personalized flashcards after completing a quiz based on the questions I struggled with, So that I can focus on reviewing areas where I need more practice and retain the material better.

**Acceptance Criteria:**

1. After completing a quiz, the system automatically generates flashcards based on questions the user answered incorrectly.
2. The flashcards are personalized and focused on the most challenging topics for the user.
3. Flashcards are displayed in an easy-to-use format, and users can review them immediately after the quiz or later.

## Validation:

Inquire interviewees as to the extent to which these use cases adequately describe problem

**Mahta:** The adjustable time limits are a great idea, especially for challenging questions. Real-time feedback will really help me understand where I went wrong instantly.

**Alex:** I like the idea of adjusting time limits per question and real-time feedback. Both of these would significantly reduce the stress I feel during timed quizzes. The addition of flashcards is an excellent feature—it would help me focus on reviewing the areas I struggle with, reinforcing my understanding after each quiz.

**Vania:** Current quiz tools lack flexibility, as multiple-choice questions often fail to address complex learning objectives. Real-time feedback is essential for understanding mistakes and seeking clarification immediately. Detailed feedback on wrong answers would greatly improve my learning process.

**Bonnie:** Typing math equations in quizzes is often challenging and can make questions look confusing, especially with concepts like indices. Providing clear explanations of answers is crucial, as it helps identify mistakes and shows areas for improvement.

## Interface Design:

### 1. Real-Time Feedback

**POST** /sessions/{sessionId}/answer/{playerId}

Submit an answer to a question during a quiz and receive immediate feedback.

#### Description:

Allows players to submit their answers and get instant feedback, including whether the answer is correct, detailed explanations, and updated scores.

#### Parameters:

| Name             | Type   | Description                              |
|------------------|--------|------------------------------------------|
| sessionId (path) | number | Unique ID of the session                 |
| playerId (path)  | number | Player Id for player in the quiz session |
| body (body)      | body   | Submitted answer data                    |

## Request Body Example

```
{  
  "questionId": 201,  
  "answerId": 301,  
}
```

| Code | Description                                           | Example Value                                                                                                |
|------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 200  | Answer submitted successfully with real-time feedback | <pre>{<br/>  "correct": true,<br/>  "detailedFeedback": "Correct! Great job.",<br/>  "score": 10<br/>}</pre> |
| 400  | Invalid input data or missing parameters              | <pre>{<br/>  "error": "Invalid questionId or answerId."<br/>}</pre>                                          |
| 401  | PlayerId is invalid or empty                          | <pre>{<br/>  "error": "Given player does not exist in session"<br/>}</pre>                                   |

## 2. Flashcard Generation

**GET** /session/{sessionId}/flashcards/{playerId}

Generate personalized flashcards based on quiz performance.

### Description:

Automatically create flashcards for the questions a user answered incorrectly during the quiz.

**Parameters:**

| Name             | Type   | Description              |
|------------------|--------|--------------------------|
| sessionId (path) | number | Unique ID of the session |
| playerId (path)  | string | playerId of player       |

**Responses:**

| Code | Description                                                                          | Example Value                                                                                                                                                                                                                                                              |
|------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | Flashcards generated successfully                                                    | <pre>{   "flashcards": [     {       "question": "What is 2+2?",       "detailedFeedback": "If you have two apples and you get two more, how many do you have? Well, 4!",       "correctAnswer": "4",       "relatedResources": ["https://math.com/sum"]     }   ] }</pre> |
| 400  | Session invalid. If session still hasn't gone through all questions or doesn't exist | <pre>{   "error": "Session does not exist " }</pre>                                                                                                                                                                                                                        |
| 401  | PlayerId invalid                                                                     | <pre>{   "error": "Given playerId did not take part in given session quiz" }</pre>                                                                                                                                                                                         |

**3. Adding a New Question Type**

**POST** /quizzes/{quizId}/questions/type

Add a new question with customizable types.

## Description

Allows quiz creators to add diverse question types like matching, short answer, true/false, etc.

## Parameters:

| Name           | Type   | Description                     |
|----------------|--------|---------------------------------|
| quizId (path)  | number | Unique ID of the quiz           |
| token (header) | string | Token of the authenticated user |
| body (body)    | body   | Submitted answer data           |

## Request Body Example

```
{
  "question": "Match the following countries with their capitals.",
  "type": "matching",
  "timeLimit": 60,
  "points": 15,
  "answerOptions": {
    setOneOptions: [ "France", "Germany", "Italy" ],
    setTwoOptions: [ "Paris", "Rome", "Berlin" ],
  },
  correctAnswers: [
    {setOne: "France", setTwo: "Paris"}
  ]
}
```

## Responses:

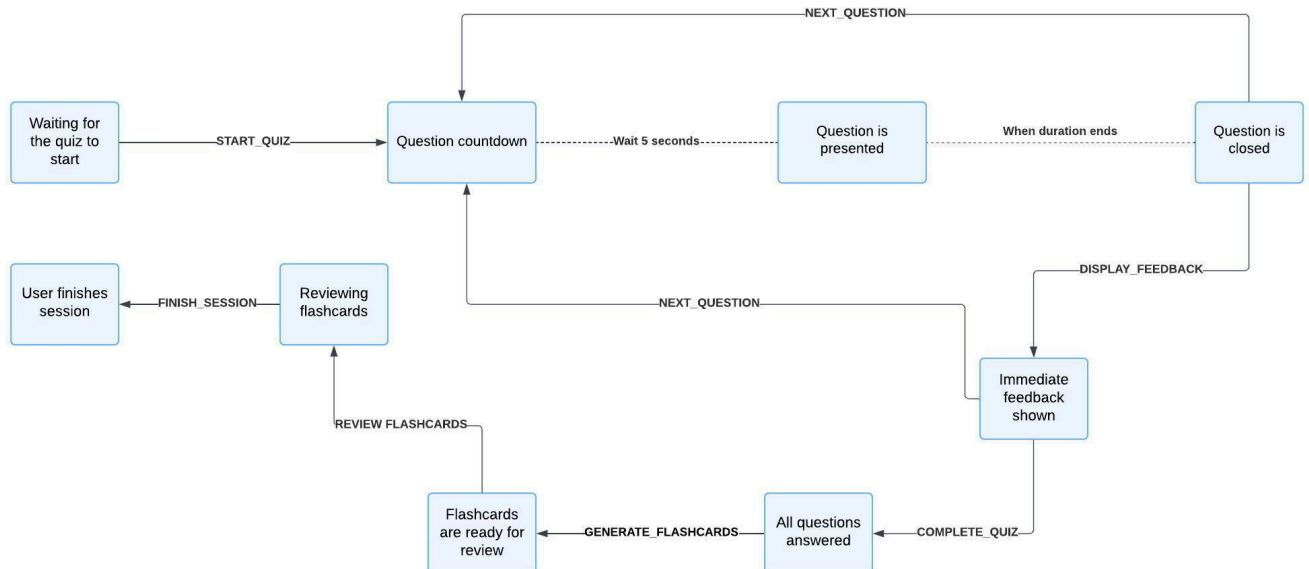
| Code | Description                              | Example Value                                                                      |
|------|------------------------------------------|------------------------------------------------------------------------------------|
| 200  | Question added successfully              | {<br>"questionId": 501,<br>"message": "Matching question added successfully."<br>} |
| 400  | Invalid input data or missing parameters | {<br>"error": "Invalid question type or missing"                                   |

|     |                           |                                                  |
|-----|---------------------------|--------------------------------------------------|
|     |                           | fields."<br>}                                    |
| 401 | Token is invalid or empty | {<br>"error": " token is empty or invalid."<br>} |

## Conceptual Modelling- State Diagrams:

[flashcards state diagram](#)

### Flashcard State Diagram:



## different question type state diagram

### Different question types State Diagram:

